Hosting Capacity and the Data-Center Connection Game

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An unprecedented surge in demand for electricity

- **Scale.** ERCOT is tracking more than 438 GW of connection requests from large loads as of mid-2026, nearly 90% of them data centers, against an all-time peak demand of 85.5 GW set on August 2023.[1]
- **Single sites are now GW-sized.**
  - OpenAI's Stargate: nearly 7 GW planned across its announced sites.[2]
  - **ANTHROP\C**: $50B of new data centers in Texas and New York;[3] up to 5 GW of compute with Amazon;[4] single training sites of 2 GW (2027) and 5 GW (2028) expected.[5]

[1] [1] ERCOT, *Trending Topics: ERCOT's New Batch Connection Process for Large Electricity Users*, 2026.

[2] [2] OpenAI, *OpenAI, Oracle, and SoftBank Expand Stargate with Five New AI Data Center Sites*, 2025.

[3] [3] Anthropic, *Anthropic Invests $50 Billion in American AI Infrastructure*, 2025.

[4] [4] Anthropic, *Anthropic and Amazon Expand Collaboration for up to 5 Gigawatts of New Compute*, 2026.

[5] [5] Anthropic, *Build AI in America: Anthropic Energy Report*, 2025.

Who decides, and the two questions for today

- In ERCOT, the transmission service provider (e.g., Oncor) and ERCOT study each large-load request.
- Since 2026, ERCOT studies eligible large loads *together* and allocates transmission capacity "based on what the grid can reliably support".[6]

1. How much load can a bus take? — Part I: hosting capacity
2. When requests compete, who gets served? — Part II: the connection game

[6] [1] ERCOT, *Trending Topics: ERCOT's New Batch Connection Process for Large Electricity Users*, 2026.

Part I: Hosting capacity

What is hosting capacity?

Hosting capacity of a location

The largest amount of new *load* that can be connected there without violating the network's operating limits, with the existing network and controls.

- The idea started with *distributed generation*, e.g. rooftop solar photovoltaic (PV) panels on distribution feeders: how much PV can a feeder absorb?[7]
- The same question applies to large new consumption,[8] and to transmission-level loads such as data centers.[9]
- Computed by stochastic simulation, streamlined rules, time series, or optimization (today).[10]

[7] [6] Bollen et al., *Integration of Distributed Generation in the Power System*, 2011, [7] Rylander et al., EPRI, 2015.

[8] [8] Bollen et al., *Energies*, 2017.

[9] [9] Cicilio et al., *IEEE Transactions on Sustainable Energy*, 2021, [10] Wan et al., arXiv:2610.18405, 2026.

[10] [11] Mulenga et al., *International Journal of Electrical Power & Energy Systems*, 2020, [12] Ismael et al., *Renewable Energy*, 2019.

Single-bus hosting capacity as a linear program

Extra load L at bus k in hour t; existing generators redispatch:[11]

$$\mathrm{HC}_{k,t} = \max_{L,\,\mathbf{p}^{\mathrm{g}},\,\mathbf{p},\,\mathbf{f}} \; L$$

$$\text{s.t.} \quad \underbrace{\mathbf{p} = M\,\mathbf{p}^{\mathrm{g}} - \mathbf{d}_t - L\,\mathbf{e}_k}_{\text{net injections (new load at } k)}, \qquad \underbrace{\mathbf{1}^{\top}\mathbf{p} = 0}_{\text{system balance}}, \qquad \underbrace{\mathbf{f} = \Phi\,\mathbf{p}}_{\text{DC power flow}}$$

$$\underbrace{|\mathbf{f}| \le \mathbf{F}}_{\text{line thermal limits}}, \qquad \underbrace{\mathbf{p}^{\min} \le \mathbf{p}^{\mathrm{g}} \le \mathbf{p}^{\max}}_{\text{generator limits}},$$

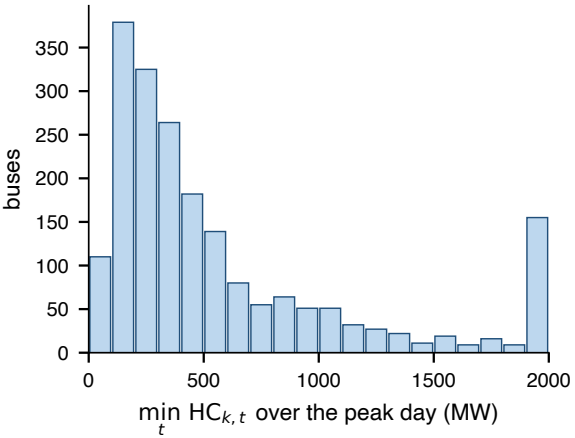
$$L \ge 0$$

Decision variables: L new load, $\mathbf{p}^{\mathrm{g}}$ generator outputs, $\mathbf{p}$ net bus injections, $\mathbf{f}$ line flows. Data: M maps generators to buses, $\mathbf{d}_t$ bus loads, Φ the PTDF (shift-factor) matrix, $\mathbf{F}$ line limits.

Assumptions: lossless DC power flow (no voltage or reactive-power limits), and the set of committed generators is fixed ($\mathbf{p}^{\min}$ is given).

[11] [13] Alturki et al., *Applied Energy*, 2018, [10] Wan et al., arXiv:2610.18405, 2026.

Hosting capacity varies a lot across buses

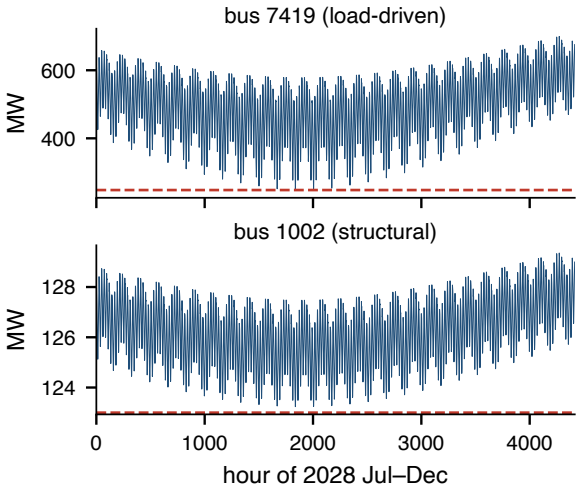


ACTIVSg2000 synthetic Texas grid,[a] 2028 Jul–Dec, the 24 hours of the peak-load day, all 2,000 buses:

- median 372 MW
- 10% of buses below 130 MW
- 90% below 1,536 MW; 146 buses above 2,000 MW (not shown)
- smallest: 10 MW

[a] [14] Birchfield et al., *IEEE Transactions on Power Systems*, 2017.

Over time: the stressed hours bind



A constant load must be served in *every* hour:

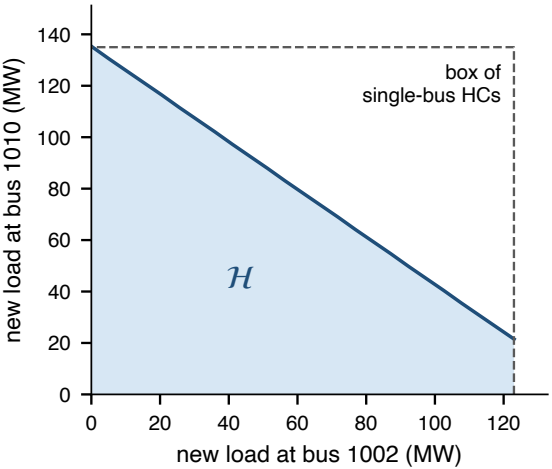
$$\mathrm{HC}_k = \min_t \mathrm{HC}_{k,t} \quad \text{(dashed)}$$

- Bus 7419: 248 MW to 699 MW, median 503 MW: tight at peak hours.
- Bus 1002: 123 MW to 129 MW: the network limits it, not the load cycle.

"Dynamic hosting capacity" treats this time variation explicitly.[a]

[a] [15] Oliveira et al., *Energies*, 2019.

Many buses at once: the hosting-capacity region



The hosting-capacity region $\mathcal{H}$: every combination of new loads $\mathbf{x}$ (one per bus) that the grid can serve in every hour. (Formal definition in the backup.)

- A convex polytope: the projection of the DCOPF feasible set.[a]
- Single-bus HCs are its axis intercepts; they do not add up.[b]
- Peak hour: bus 1002 hosts 123 MW alone, only 81 MW with 60 MW at neighbor 1010.

[a] [16] Gong et al., *Electric Power Systems Research*, 2024.
[b] [10] Wan et al., arXiv:2610.18405, 2026.

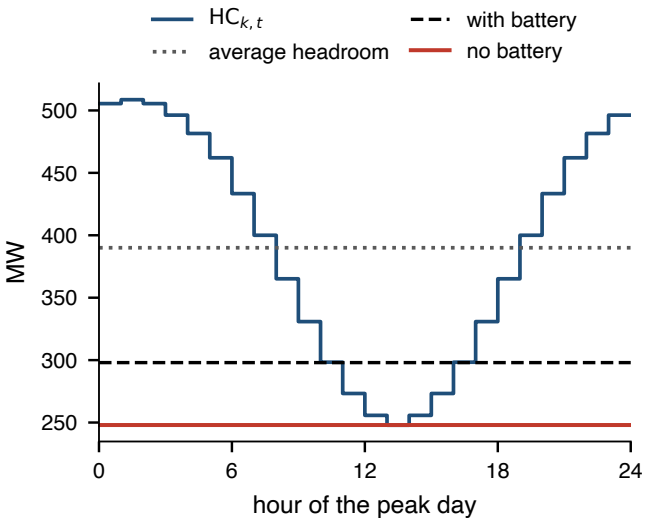
With a local generator

Site at bus k: load L, private generator $0 \le g_t \le G$. The grid only sees the *net* withdrawal $L - g_t$:

$$\text{feasible} \iff \exists\, g_t \in [0, G] : \underbrace{L - g_t}_{\text{net withdrawal}} \le \mathrm{HC}_{k,t} \ \ \forall t \implies \boxed{\mathrm{HC}_k^G = \min_t \mathrm{HC}_{k,t} + G}$$

- Exact when the generator is always available; with outages, ramping, or fuel limits, $\mathrm{HC}_k^G \le \mathrm{HC}_k + G$.
- The generator runs only when the grid is short or when its cost is below the local price.

Storage: move energy from slack hours to tight hours



Bus 7419, peak day, hourly steps, daily cycle ($e_0 = e_{24} = B/2$), $B = 4P$:

- no battery: 248 MW
- average headroom (dotted): 390 MW, the most any battery could reach (next slide)
- $P = 50$ MW, $B = 200$ MWh: 298 MW

B limits how long a shortfall the battery can cover. Half-year: 248 $\rightarrow$ 298 MW.

With storage: the hours become coupled

Battery at bus k: power P, energy B, signed output u_t (positive = discharge), time step Δt; $\mathrm{HC}_{k,t}$ comes from the single-bus LP:

$$\mathrm{HC}_k^S = \max_{L,\,u,\,e} \; L \quad \text{s.t.} \qquad \underbrace{L - u_t \le \mathrm{HC}_{k,t}}_{\text{grid limit on net withdrawal}}, \qquad \underbrace{-P \le u_t \le P}_{\text{power}},$$

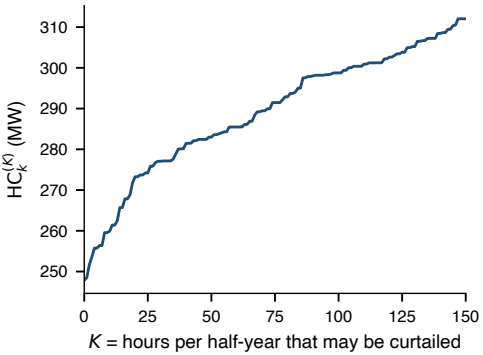
$$\underbrace{e_t = e_{t-1} - \Delta t\, u_t, \quad 0 \le e_t \le B}_{\text{energy}}, \qquad \underbrace{e_0 = e_T}_{\text{daily cycle}}.$$

Decision variables: L new load, u_t battery output, e_t stored energy. A lossless battery (no round-trip losses), which makes the bound below optimistic.

- A multi-period LP: discharge in tight hours, recharge in slack ones.[12]
- Summing over the cycle ($\sum_t u_t = 0$): $\mathrm{HC}_k^S \le \frac{1}{T}\sum_t \mathrm{HC}_{k,t}$: storage lifts the limit at most from the minimum to the average headroom; power ($L \le \mathrm{HC}_{k,t} + P$) and energy ($B$) keep it below that.

[12] [17] Macedo et al., *IEEE PES Innovative Smart Grid Technologies Conference (ISGT)*, 2017.

Flexibility: curtail in a few hours



Flexible connection:[a]

$$\max_{L,\,\delta} \; L$$

$$\text{s.t. } L - \delta_t \le \mathrm{HC}_{k,t}, \quad \delta \ge 0, \quad \|\delta\|_0 \le K$$

- δ_t is the load curtailed in hour t; nonconvex ($\|\delta\|_0$ counts curtailed hours).
- $\mathrm{HC}_k^{(K)} = \mathrm{HC}_{k,[K+1]}$, the $(K+1)$-th *smallest* hourly value.
- Bus 7419: 248 MW at $K = 0$, 274 MW at $K = 24$ (+11%), 299 MW at $K = 100$ (+21%). Diminishing returns in K.[b]

[a] [18] Gu et al., *Hawaii International Conference on System Sciences (HICSS)*, 2026.

[b] [19] Etherden et al., *IEEE PowerTech*, 2011, [20] Sarker et al., arXiv:2604.20185, 2026.

How much should we trust a hosting-capacity number?

- **Security.** Dropping N-1 contingencies overstates data-center hosting capacity by about 20% on the IEEE RTS-24.[13]
- **Dynamics.** The data center's own ride-through limits (UPS, converters) can be the binding constraint.[14]
- **Model.** DC power flow has no voltage or reactive-power limits and no losses.
- **Simultaneity.** Single-bus numbers ignore everyone else who is connecting at the same time.

The last point leads to the second question: who gets served?

[13] [10] Wan et al., arXiv:2610.18405, 2026.

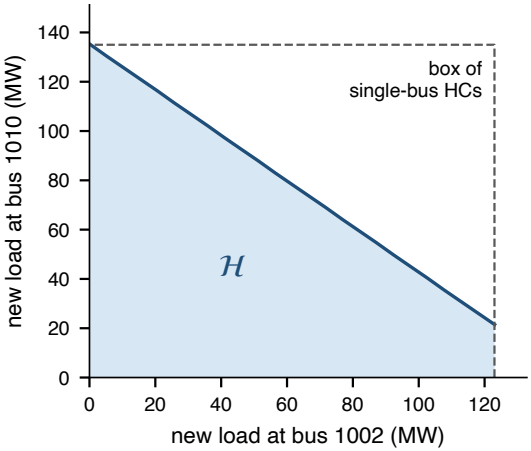
[14] [21] Ren et al., arXiv:2609.03030, 2026.

Part II: The connection game

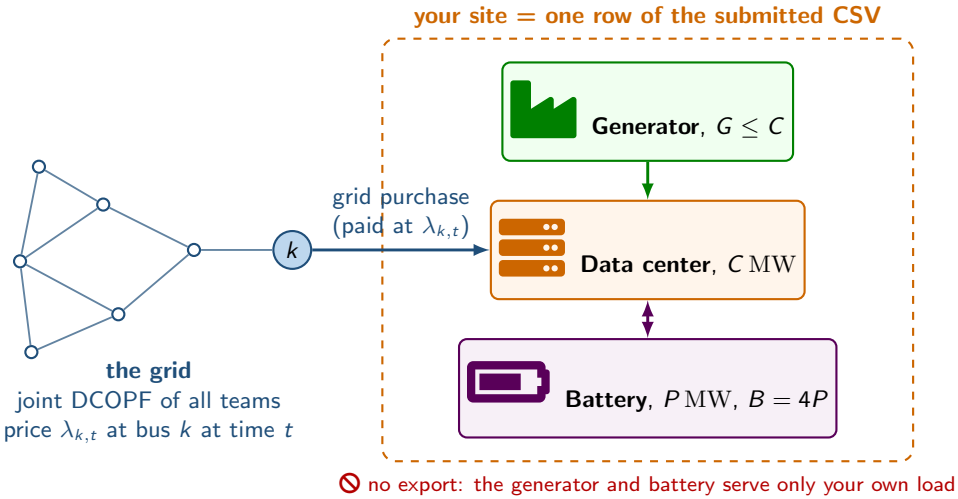
From one bus to many applicants

- Part I: one bus $\Rightarrow$ one number; many buses $\Rightarrow$ a region.
- Two requests can each fit alone and still not fit together.
- Someone has to decide who is served, and by how much.

For the next three weeks, you are the applicants.



Your role in one picture

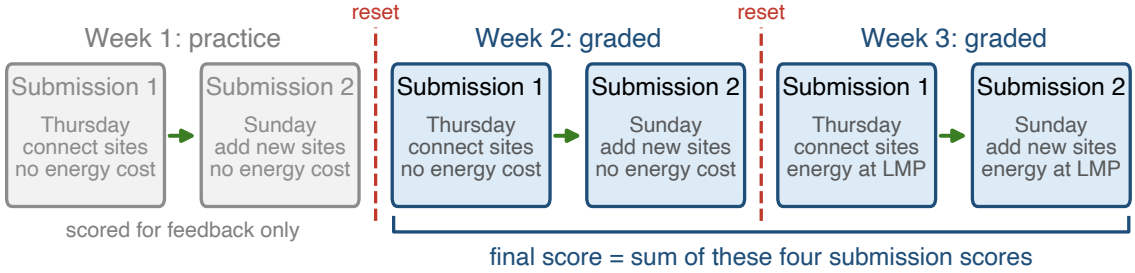


You choose the numbers C, G, P for each site, and the grid operator runs them.

What earns money, and when

One 50 MW site, planned on 5 representative days (120 hours) for 10 years:

- earns $1,000 per MWh served over ten years (7% discount), $61.5M per MW; costs $25M per MW (paid once).
- Energy is about $18/MWh, charged only in Week 3.
- It breaks even at about 41% served, so what matters is whether the grid can deliver.



What the grid operator does

Part I's LP, with every team's submitted plans:

$$
\begin{aligned}
&\min \text{ generation cost} + \text{private generator cost} \\
&\text{s.t. } \mathbf{p} = M\mathbf{p}^{g} - \mathbf{d}_t - (\text{all teams' sites, net of their own generator and battery}), \\
&\quad \mathbf{1}^{\top}\mathbf{p} = 0, \quad \mathbf{f} = \Phi\mathbf{p}, \quad |\mathbf{f}| \le \mathbf{F}, \quad \text{batteries' constraints.}
\end{aligned}
$$

- One DCOPF per representative hour (120 in all), coupled through the batteries over each day.
- The price of energy at your bus is the LMP.

When it doesn't fit: how ERCOT connects large loads since 2026

Before: one request at a time

- Each new project could invalidate the studies of projects ahead of it.
- Repeated restudies; delays of years.

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Now: a batch (PGRR145, "Batch Zero")[a]

- All eligible loads ($\geq 75\,\mathrm{MW}$) are studied together.
- MW are allocated "based on what the grid can reliably support".

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- **Base load:** studied before and its MW are kept.
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Minimum allocation[a]

- Request above $100\,\mathrm{MW}$: an allocation below $100\,\mathrm{MW}$ is reduced to zero.
- Request of $100\,\mathrm{MW}$ or less: an allocation below 90% of the request is reduced to zero.

Ways to connect sooner

- Bring your own generation, or let ERCOT curtail you under stress (Part I).

[a] [22] ERCOT, ERCOT Board of Directors, Item 15.1.4.3, 2026.

The game's batch rule: one shared limit

Three teams behind one line that can carry $200\,\mathrm{MW}$ of new load.
Requests: A 40, B 100, C 200 (total 340).

	request	step 1	share
A	40	40	100%
B	100	80	80%
C	200	80	40%

[15] [23] Sprumont, *Econometrica*, 1991.

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3. **Recompute without C.** $40 + 100 = 140 \leq 200$: A and B are served in full.

	request	final	share
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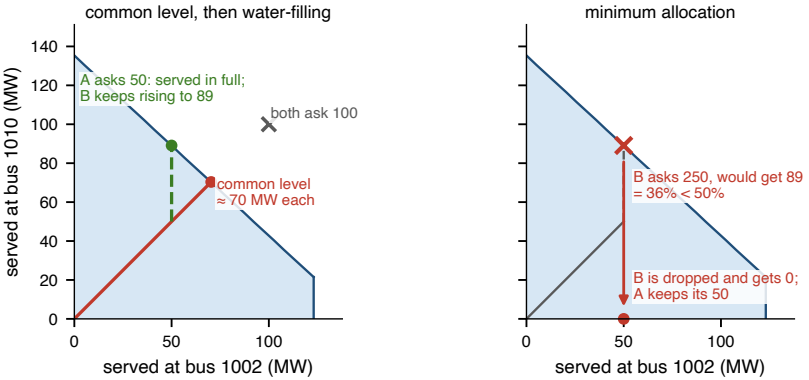
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C still pays capital on 200 MW. Asking for more than the grid can give is the most expensive mistake in the game.

ERCOT does not publish a formula for splitting capacity among studied loads. The game uses equal MW (the uniform rule[15]) so that everyone can predict it.

[15] [23] Sprumont, *Econometrica*, 1991.

On the network: sharing the region



1. Everyone rises to a common MW level. A team that asked for less is served in full.
2. Teams that are not blocked keep rising (water-filling).
3. A team that would get less than half of its request gets nothing.

The game and ERCOT

	ERCOT batch study	This game
Who is studied together	All eligible loads in a batch	All teams' submissions
Already connected	Base load keeps its MW	`ACTIVSg2000`'s existing load
How MW are split	System-wide study	Equal MW level, then water-filling
Too small a share	below $100\,\mathrm{MW}$ (90% for requests $\leq 100\,\mathrm{MW}$) $\to 0$	below 50% of the request $\to 0$
Commitment	Financial security, site control	Capital charged on the full request
Own generation	Offsets transmission need	Private generator G
Flexible load	May be curtailed under stress	Not in the game (Part I)

A two-team example

Team A at bus 1002, team B at neighbor 1010; each picks 50 or 100 MW in Submission 1.

Ten-year payoffs, Weeks 1–2 ($M), A's payoff / B's payoff:

A \ B	B small (50)	B large (100)
A small (50)	1,826 / 1,826	1,826 / 2,980
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Boxed: each team's best response to the other's choice. A cell with both boxed is a Nash equilibrium.

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- Nash equilibria: (A small, B large); (A large, B small). A game of *chicken*: if both insist, both lose. In Week 3, congestion adds $400–770M of energy cost per team.

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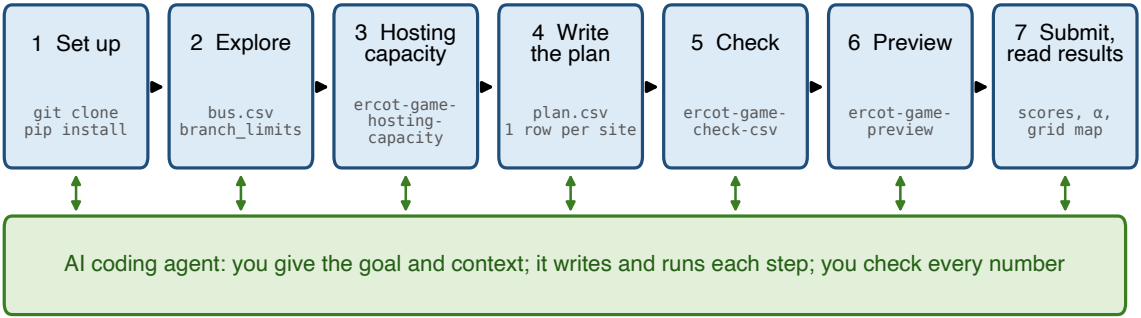
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- Nash equilibria: (A small, B large); (A large, B small). A game of *chicken*: if both insist, both lose. In Week 3, congestion adds $400–770M of energy cost per team.
- Your score depends on where everyone builds, and you see the results between submissions. What would you do?

From the slides to your laptop



Everything is in one GitHub repository: [Link].

The tutorial walks through a full session with an AI coding agent on a 14-bus demo.

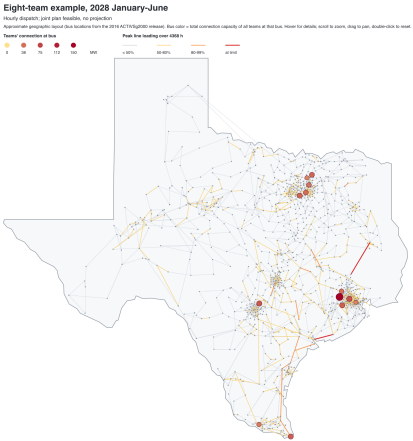
After each round, and your first plan

What you get back

- your score breakdown (in private);
- every team's service fraction $\alpha\%$;
- a grid map with the total MW at each bus.

A first plan for Week 1

1. Shortlist buses by hosting capacity.
2. Ask well below it if w/o a generator or battery.
3. Check, preview, change one thing, compare.



Eight-team example: bus color = total MW of all teams, line color = peak loading; approximate locations.

Takeaways

- Hosting capacity is an LP: the DCOPF with a "maximize the new load" objective.
- Time: a constant load is limited by the worst hour. Generation adds G and storage moves the limit toward the average. Flexibility toward the $(K+1)$-th worst.
- Space: single-bus numbers are intercepts of a region and do not add up.
- With many applicants, who gets served is a design choice, and it turns planning into a game.

Questions for you. How should a grid operator allocate scarce headroom among data centers? What would you change in the game's rule?

References I

[1] ERCOT. *Trending Topics: ERCOT's New Batch Connection Process for Large Electricity Users*. June 2026.

[2] OpenAI. *OpenAI, Oracle, and SoftBank Expand Stargate with Five New AI Data Center Sites*. Sept. 2025.

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[6] M. H. J. Bollen and F. Hassan. *Integration of Distributed Generation in the Power System*. Wiley-IEEE Press, 2011.

[7] M. Rylander, L. Rogers, and J. Smith. *Distribution Feeder Hosting Capacity: What Matters When Planning for DER?* Tech. rep. 3002004777. EPRI, 2015.

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Backup: the evaluator in full

For each representative day $d = 1, \dots, 5$ and hour $t = 1, \dots, 24$ ($\Delta t = 1$ h, 120 hours in all): a DCOPF with all teams' sites, coupled within each day through batteries and generator run-hour budgets.

$$\min \ \sum_{d,t} \Delta t \big(\mathbf{c}^\top \mathbf{p}^{\mathrm{g}}_{dt} + \sum_s \sum_m c^m q^m_{s,dt} \big)$$

s.t. DCOPF balance and line limits with loads $\alpha_i C_s$ and private q_s, u_s

$$e_{s,d,t} = e_{s,d,t-1} - \Delta t \, u_{s,dt}, \quad 0 \le e \le B_s, \quad |u| \le P_s, \quad e_{s,d,0} = e_{s,d,24} = B_s/2$$

$$\sum_{t,m} \Delta t \, q^m_{s,dt} \le h \, G_s \quad \text{(generator run-hour budget, } h = 2 \text{ h per day)}$$

$$\sum_{s \in i,\, k(s)=n} u_{s,dt} \le \sum_{s \in i,\, k(s)=n} \alpha_i C_s \quad \text{(a team's assets serve only its own load)}$$

- An exact LP that separates by day (Benders or one LP per day).
- LMP = dual of nodal balance; it prices energy in Week 3.
- Scores weight each hour by the days it stands for, over ten years in present value.

Backup: the hosting-capacity region

The single-bus LP's constraints with new loads $\mathbf{x}$ (one per bus) in place of $L\,\mathbf{e}_k$:

$$\mathcal{H} = \left\{ \mathbf{x} \geq 0 : \begin{array}{c} \forall t \;\exists\, \mathbf{p}_t^{\mathrm{g}}, \mathbf{p}_t : \\ \mathbf{p}_t = M\,\mathbf{p}_t^{\mathrm{g}} - \mathbf{d}_t - \mathbf{x}, \quad \mathbf{1}^\top \mathbf{p}_t = 0, \\ |\Phi \mathbf{p}_t| \leq \mathbf{F}, \quad \mathbf{p}^{\min} \leq \mathbf{p}_t^{\mathrm{g}} \leq \mathbf{p}^{\max} \end{array} \right\}$$

A convex polytope: the projection of the DCOPF feasible set onto the new loads. The single-bus hosting capacity HC_k is its intercept on axis k.